

NLP Quiz 11

NAME :

BID :

NOTE: Please have your writing readable (PRINT). Any unreadable content won't be regarded at the grader's discretion.

- 1 Consider a linear regression model $w^T x$. What's the size of hypothesis set for case $x \in R^1$? And what's the size of hypothesis set for case x is binary? (i.e., x is either 0 or 1)

- 2 In attention design, we have to create a query to key attention matrix. If I have an input sequence with length 5, with embedding size 100, how many entries will the attention matrix have? How many entries will the attention matrix have if now I have length 10 with embedding size 100? How about now I keep the input length 5, but increase dimension of embedding to 200?

- 3 Describe what's the Perceptron Model (in math) (0.5pt)? Describe the condition that the Perceptron Model will not converge in 1 sentence. (1pt)

- 4 What's the intuition of designing Principal Component Analysis? (Not what it's used for)